

Experiment 8.2

Determination and Plotting of Magnetic Flux Density vs Distance

Code File Name: Exp8_2_MagneticFluxDensity.m

Code

```
clc;  
I = 5;  
mu0 = 4*pi*1e-7;  
r = linspace(0.1,10,100);  
H = I./(2*pi*r);  
B = mu0 * H;  
plot(r,B,'LineWidth',2);  
xlabel('Distance (m)');  
ylabel('Magnetic Flux Density (T)');  
title('Magnetic Flux Density vs Distance');
```

Output Graph

